

🐒 Extra — set or change? Two Ways to Give a Variable a Number

Topic: `set` vs `change` · **Course:** Builder Basics · **Level:** Intermediate · **Time:** about 35 minutes

A variable is a number with a name. `set` gives it a brand new number. `change` adds to the number it already holds.

1 · Predict 🧠

Read the code. The agent moves the number of blocks that `monkey` holds.

```
set monkey to 2
move forward by monkey
change monkey by 3
move forward by monkey
set monkey to 1
move forward by monkey
```

Fill the table. Write what `monkey` holds before each move.

Move	<code>monkey</code> before this move
1	2
2	
3	

How many blocks does the agent move in total?

2 · Which Block Do You Need?

monkey holds 5. You run **change monkey by 5** . Now **monkey** holds... Circle one: 5 · 10 · 25

monkey holds 5. You run **set monkey to 5** . Now **monkey** holds... Circle one: 5 · 10 · 0

You want the number to start at 6. Circle one: **set monkey to 6** · **change monkey by 6**

monkey holds 6 and you want 6 more. Circle one: **set monkey to 6** · **change monkey by 6**

Finish the rule in your own words.

set means ____.

change means ____.

3 · Find the Bug 🐛

Bug A — This should build three walls: 4 blocks long, then 8, then 12. It builds 4, then 12, then 12.

```
set monkey to 4
move forward by monkey
set monkey to 12
move forward by monkey
set monkey to 12
move forward by monkey
```

Hint: what number does the second wall need?

Write the fixed code. Use `change` instead.

Bug B — The first wall should be 4 blocks long. It comes out 8.

```
set monkey to 4
repeat 3 times
  change monkey by 4
  move forward by monkey
```

Which two lines swap places? Write the fixed code.

Why does the order of those two lines matter? One sentence.

4 · Count the Corner

The agent starts on **A1** facing right. It places 5 blocks along row **A**. Then it turns right and finishes the L.

	1	2	3	4	5
A	■	■	■	■	■
B					■
C					■

Count the ■ blocks in the picture. How many in total? Circle one: 7 · 8 · 10

After the turn, how many more blocks does the agent still place? Circle one: 2 · 3 · 5

The corner block is A5. Why is it not placed twice? One sentence.

Write the code for this L. Use monkey for the long arm.

5 · Show Your Work 📷 🎥

Switch to your homework world. Build **three walls in a row**: the first 4 blocks long, the second 8, the third 12. Use one variable and **change** it — no **set** inside the loop. Then change **one number** (the start value or the change amount) and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does. Point at your **change** block and say the new number it makes.

2 · Your build.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.